

The background of the slide is a dark, black sky filled with numerous small, glowing orange lanterns. These lanterns are scattered across the frame, some appearing closer and larger, while others are further away and smaller, creating a sense of depth and movement. The lanterns have a warm, yellow-orange glow from their internal lights.

Permanent Maxillary Lateral Incisors

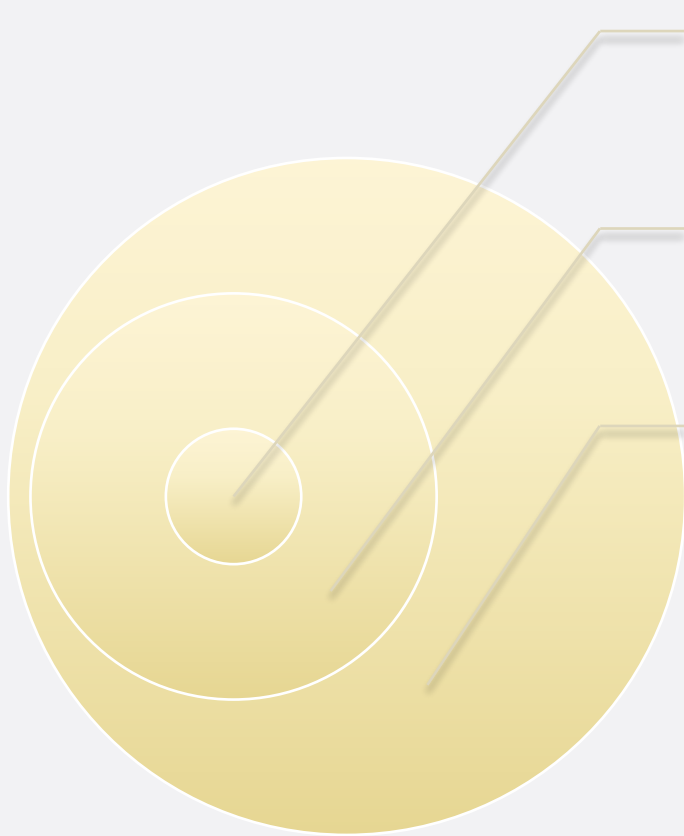
Dr Saba Arshad
Senior Lecturer

Department of Oral Biology

Learning Objectives

- To define and pronounce the terminologies of permanent dentition.
- To label the anatomical landmarks of permanent maxillary central incisor.
- To discuss the curves and segments of maxillary central incisors.
- To be able to correctly draw the graph outline.
- To be able to carve the teeth on wax models.
- To discuss anatomical variations.

Why it is important to describe every curve and segment of normal tooth



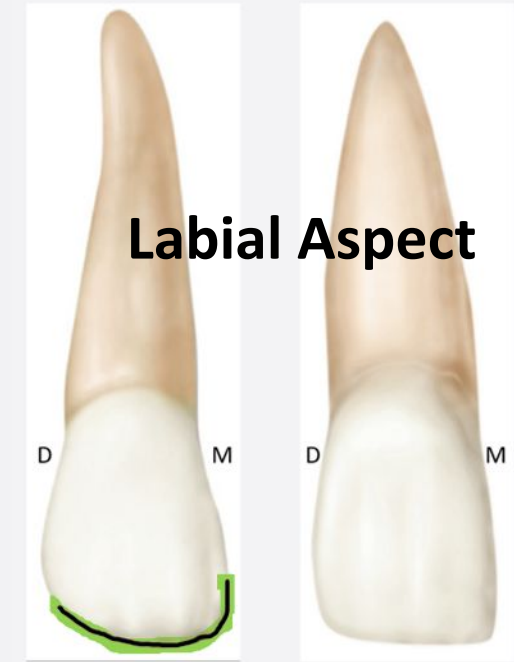
To understand the functional basis.

To establish arbitrary norm for individual teeth.

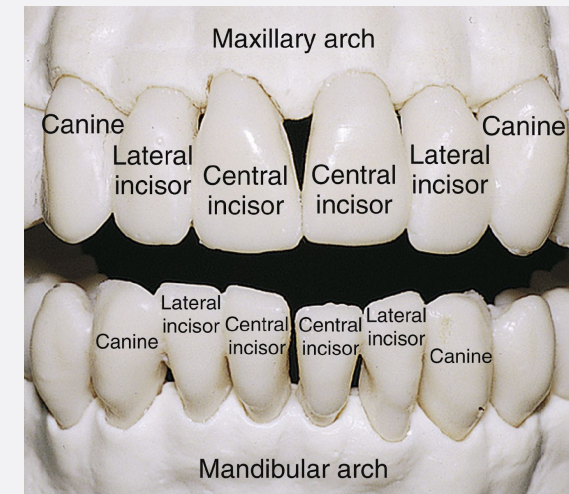
- **For comparative study of natural sizes, their normal alignment and occlusion.**
- **To reproduce the anatomy accurately to restore the defected or lost tooth form**

Maxillary Lateral Incisor – Key Features

- Assists central incisor in esthetics & incision
- **Smaller than central incisor** in all crown dimensions; MD: BL: CI
- **Root often equal to or longer than central incisor**
- **Root $\approx 1.5 \times$ crown length**
- **Crown more curved and rounded**
- **Highly variable tooth** (second only to third molars)



Maxillary lateral incisor Maxillary central incisor

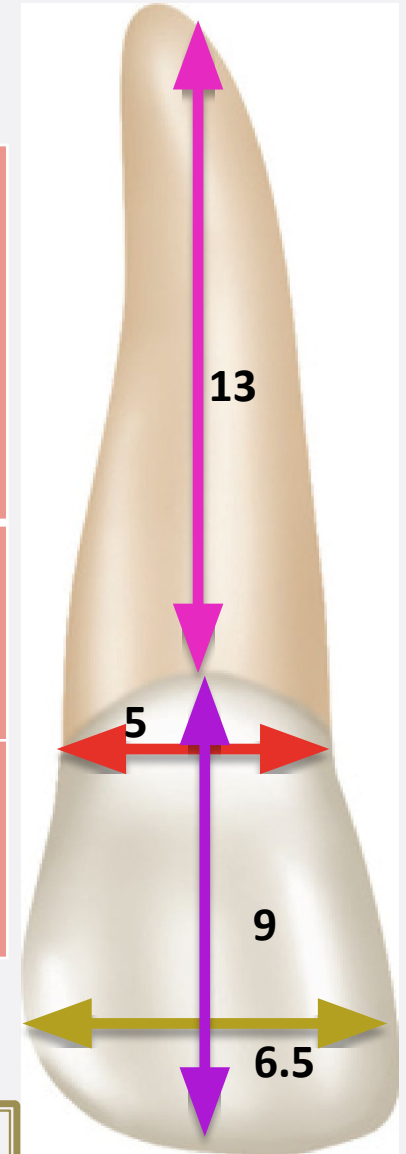


Chronology of Permanent Maxillary lateral Incisors

1st Evidence Of Calcification	10-12 Months
Enamel (Crown) Completed	4–5 Years
Eruption	8-9 Years
Root Completed	11 Years

Dimensions of Maxillary Incisors

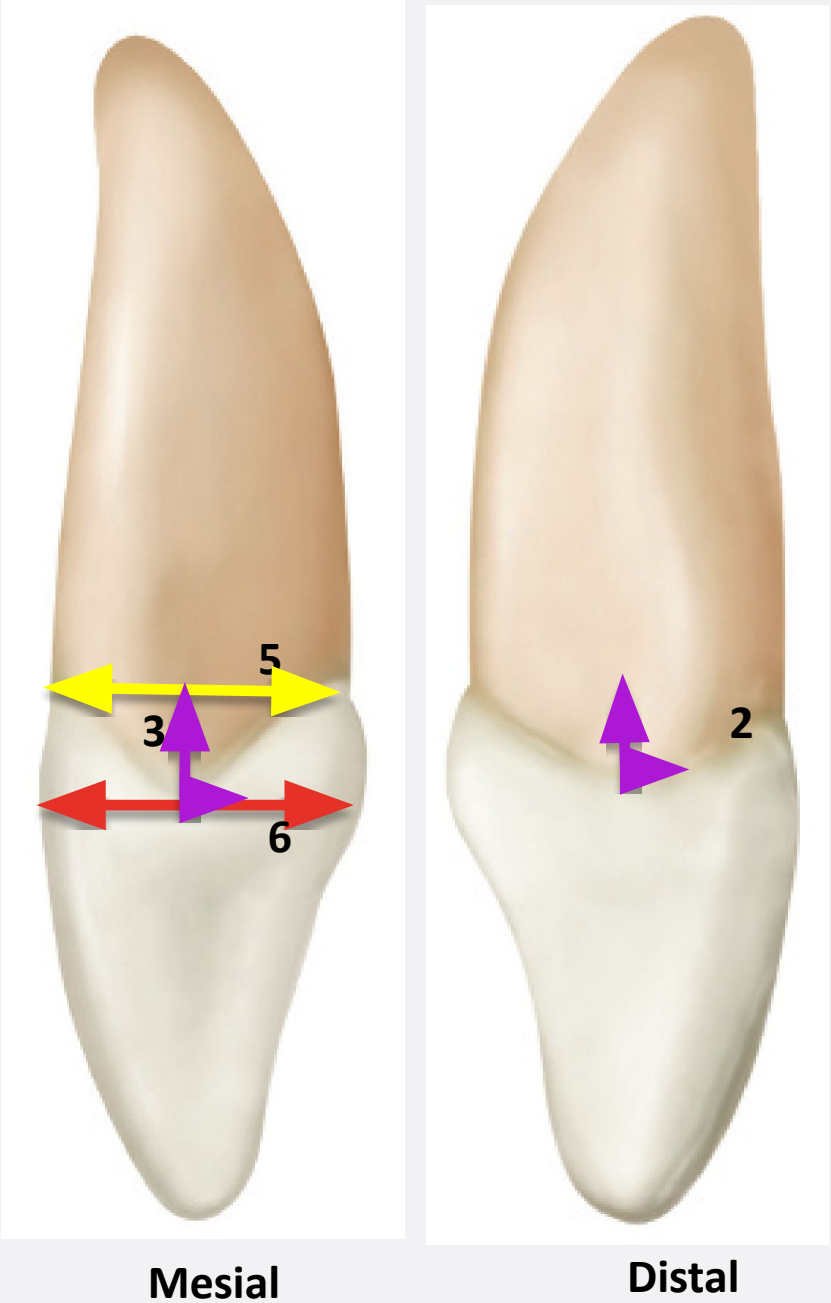
	Cervico-incisal Length of Crown	Length of Root	Mesiodistal Diameter of Crown	Mesiodistal Diameter of Crown at Cervix
Central Incisor	10.5 mm	13 mm	8.5 mm	7mm
Lateral Incisor	9 mm	13 mm	6.5 mm	5 mm



1.5 to 2mm less at cervical line than incisal edge

Dimensions of Maxillary Incisors

	Labio-lingual Diameter of Crown	Labio-lingual Diameter of Crown at Cervix	Curvature of Cervical Line—Mesial	Curvature of Cervical Line—Distal
Central Incisor	7 mm	6 mm	3.5 mm	2.5 mm
Lateral Incisor	6 mm	5 mm	3 mm	2 mm



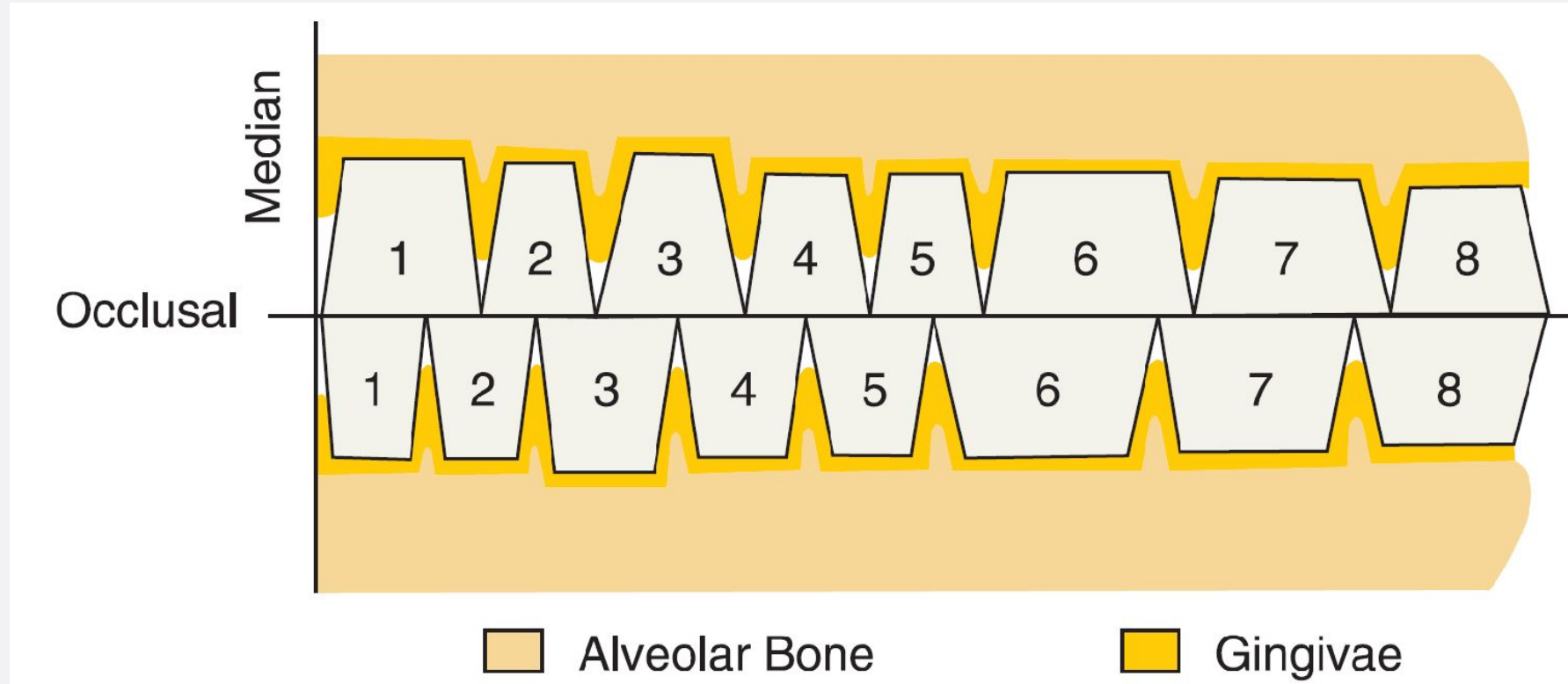
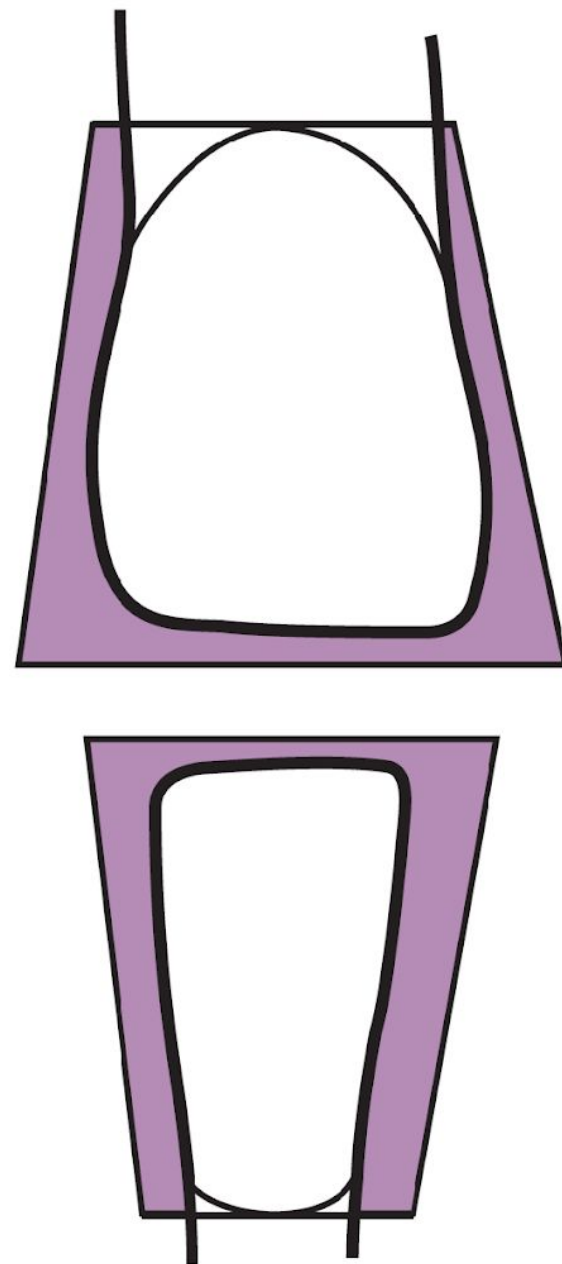
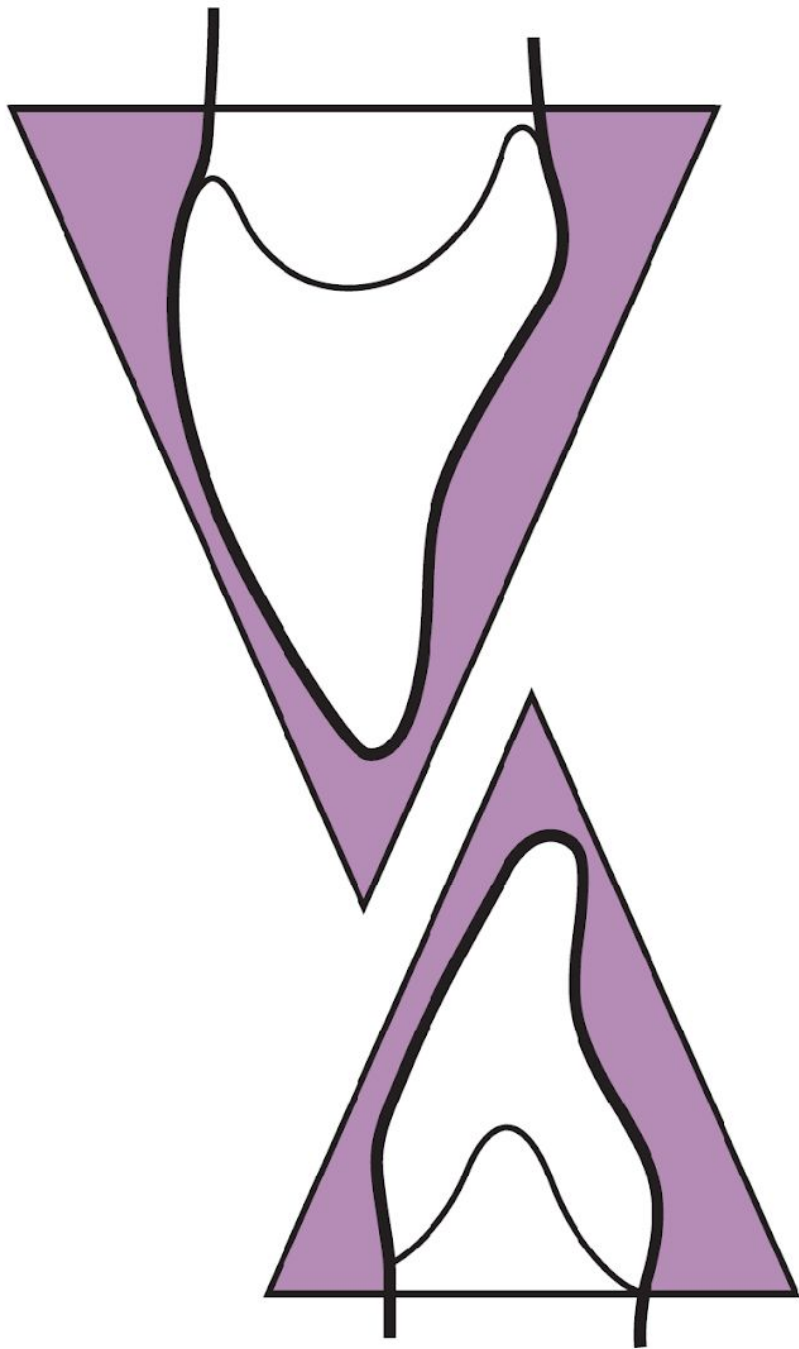
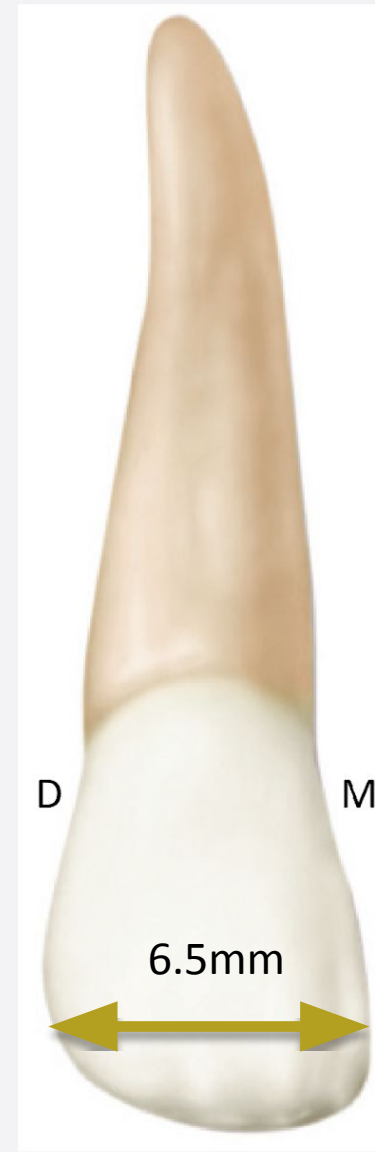
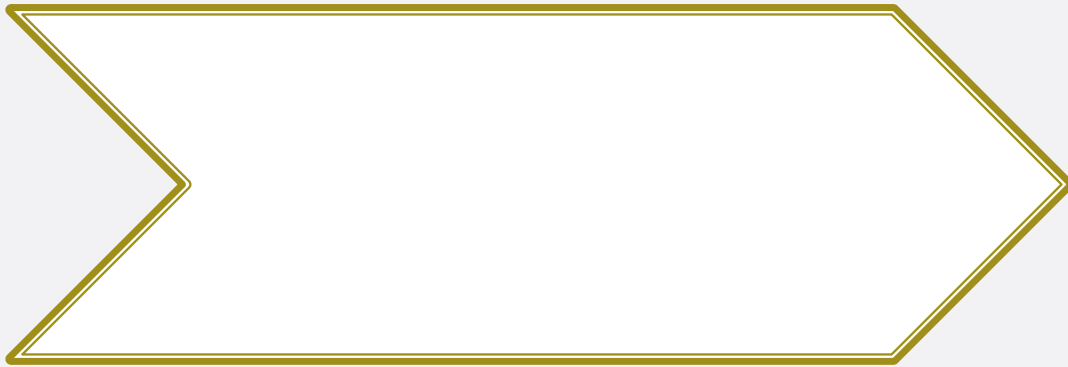


FIGURE 4-15; Trapezoidal geometric outlines of the facial and lingual aspects of tooth crowns

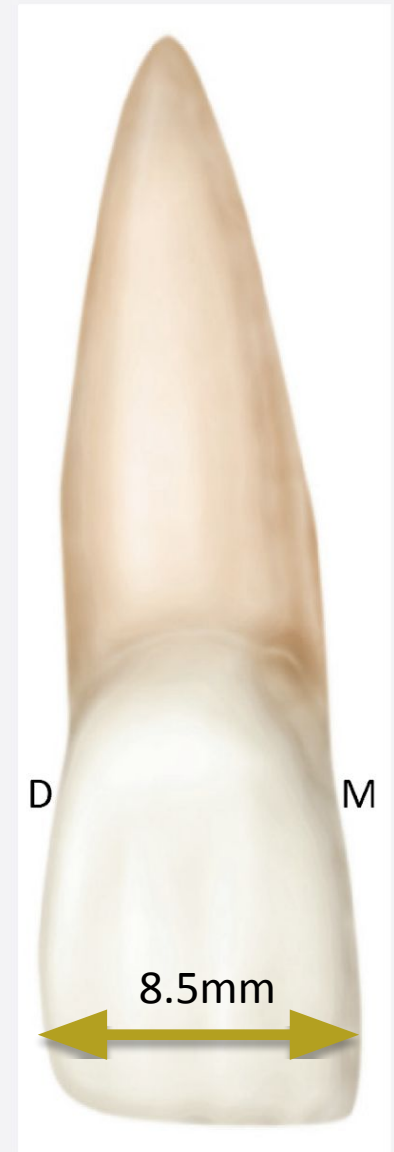
- The shortest of the uneven sides represent the bases of the crowns at the cervices,
- The longest of the uneven sides represent the working surfaces, or the incisal and occlusal surfaces.
- Each tooth has two antagonists except number 1 below and number 8 above.



Labial Aspect: Dimensions

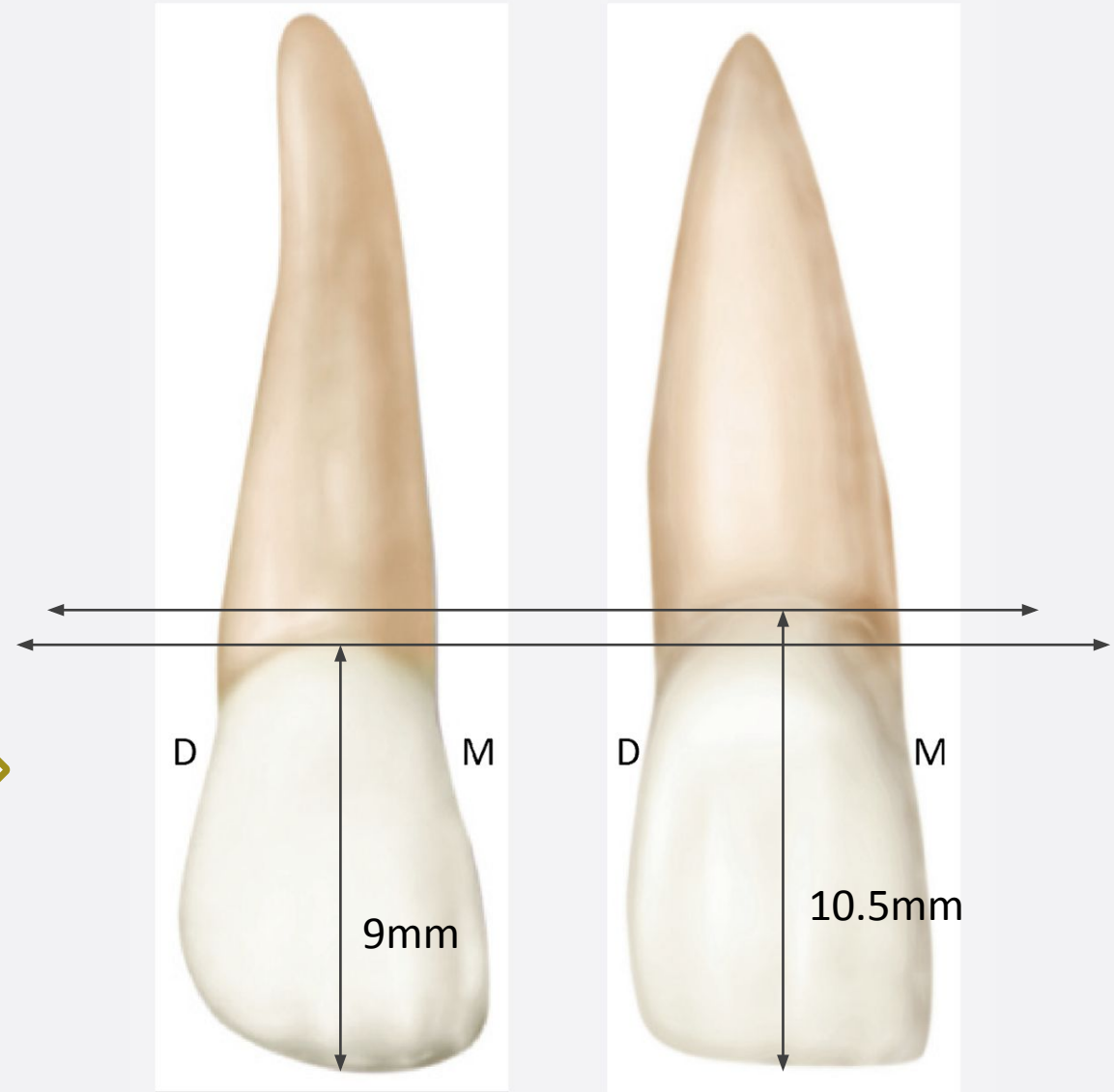


Maxillary lateral incisor



Maxillary central incisor

Labial Aspect: Dimensions

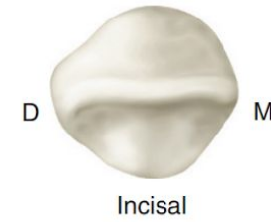


Maxillary lateral incisor Maxillary central incisor

DETAILED DESCRIPTION OF THE MAXILLARY RIGHT LATERAL INCISOR



Facial/Labial



Incisal



Lingual



Mesial



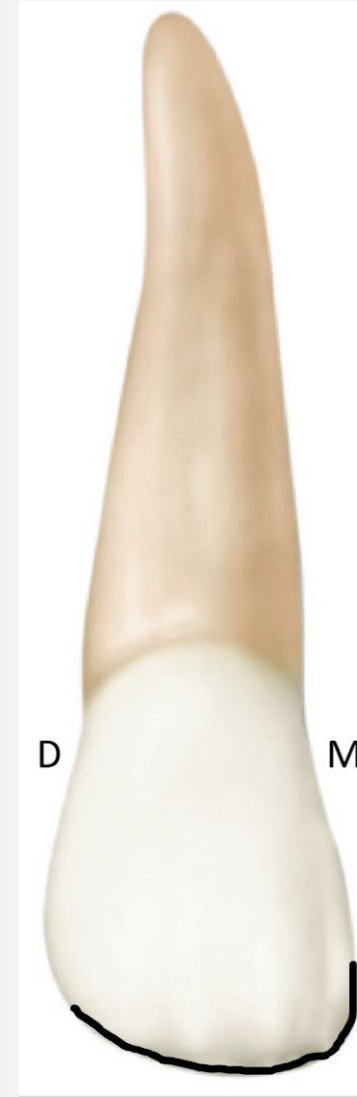
Distal

Labial Aspect

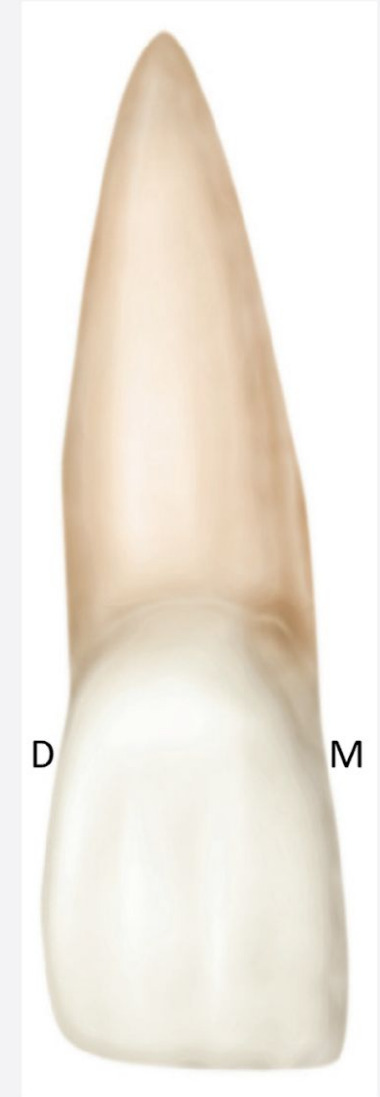
The labial surface of the crown corresponding to that of the central incisor

Overall more convex / rounded

With rounded incisal angles mesially and distally.



Maxillary lateral incisor



Maxillary central incisor

Labial Aspect

The mesial outline of the crown resembles that of central incisor

with a more rounded mesio-incisal angle.



Maxillary lateral incisor



Maxillary central incisor

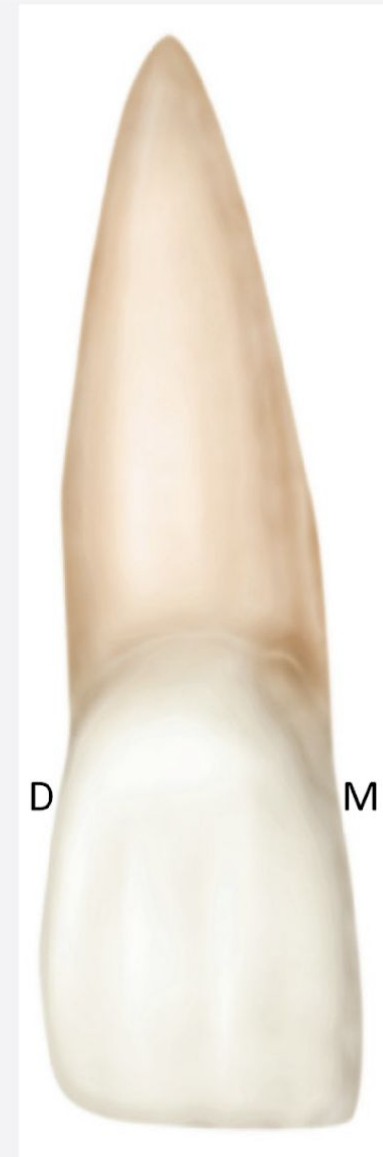
Labial Aspect

**The distal outline is
always more rounded**

**with more rounded disto-incisal
line angle/ compared to MI line
angle of lateral and to central
maxillary incisor**



Maxillary lateral incisor

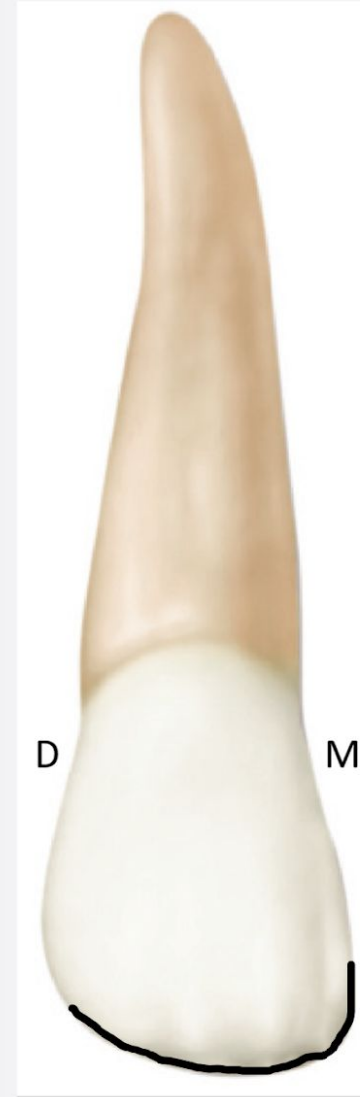


Maxillary central incisor

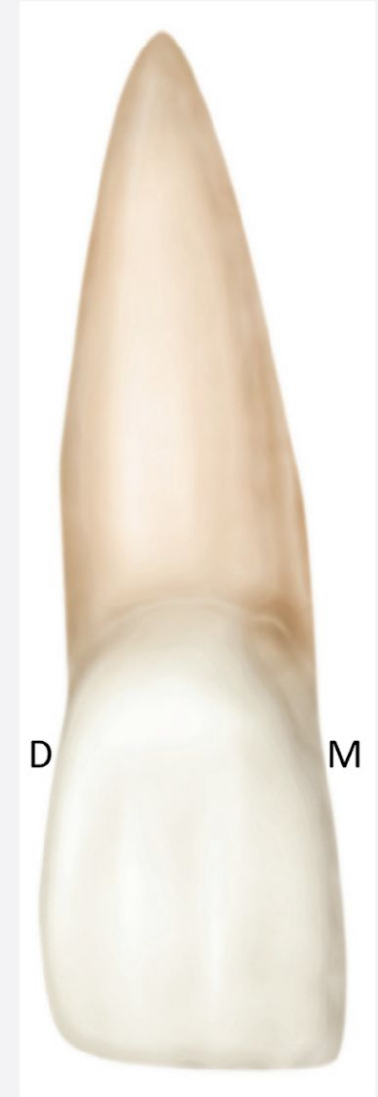
Labial Aspect

Incisal Ridge

Rounded not straight



Maxillary lateral incisor



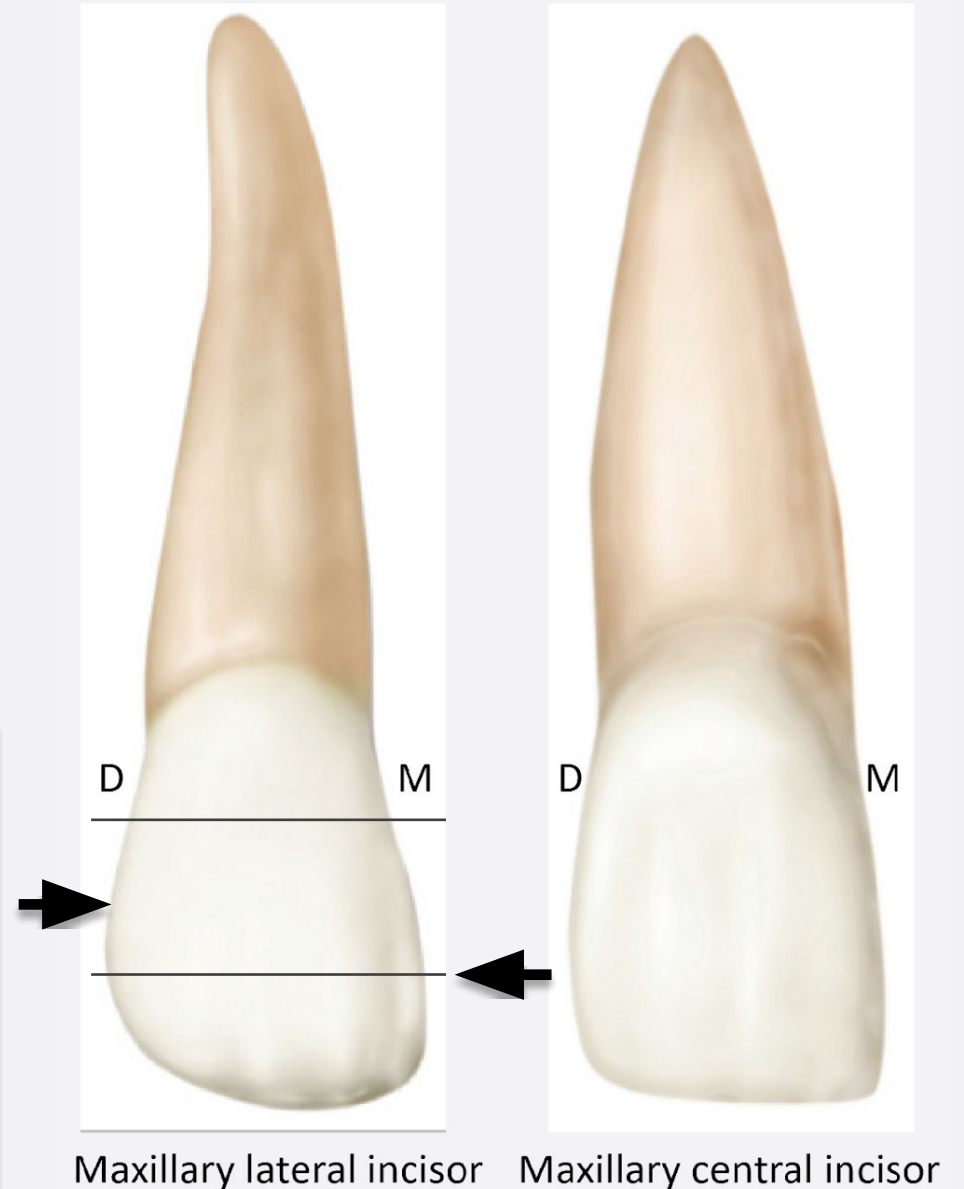
Maxillary central incisor

Labial Aspect

contact points

Mesially lies at the point of junction of the incisal and middle thirds.

Distally lies more cervical, usually in the center of the middle third.

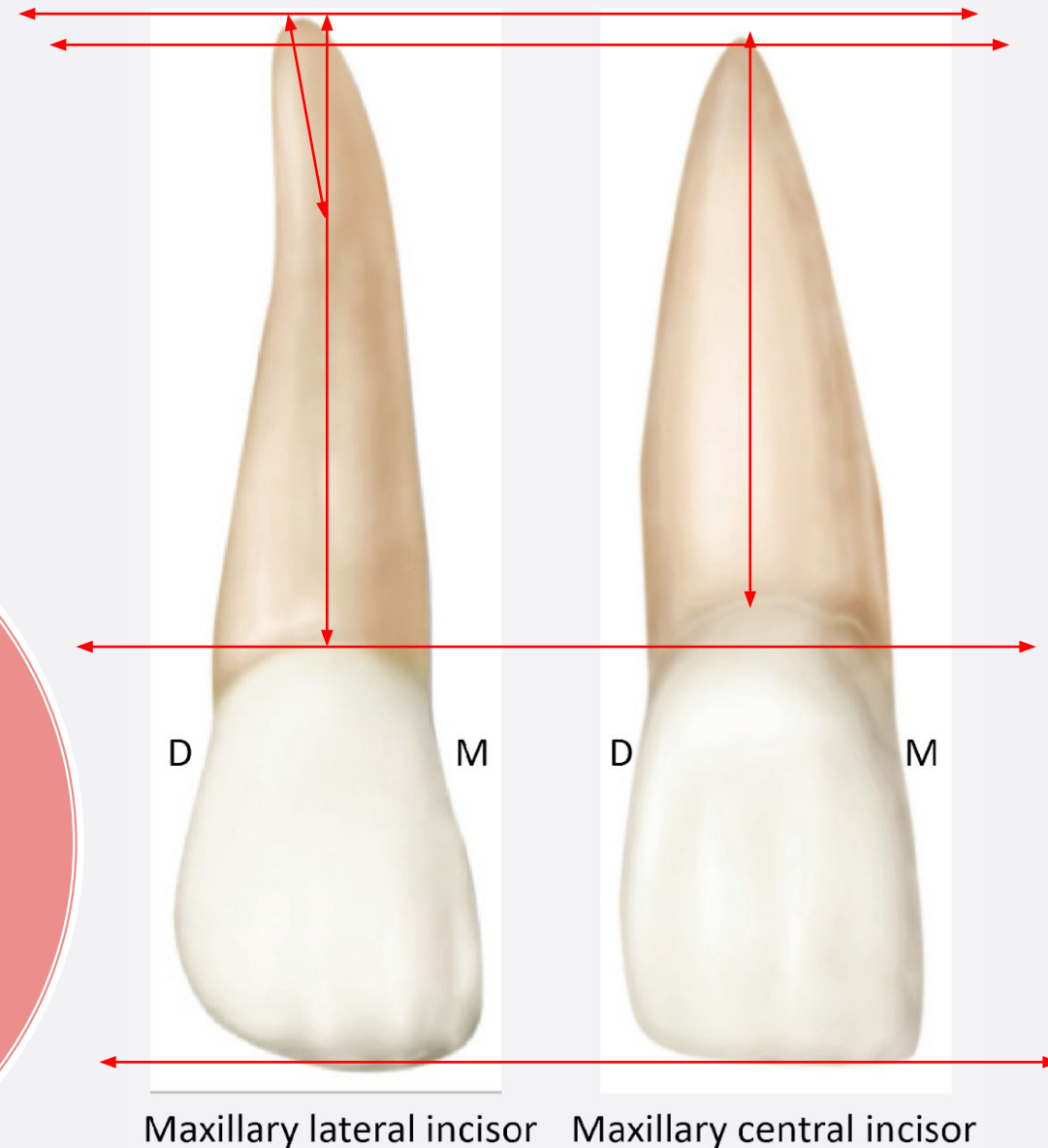


Labial Aspect: Root

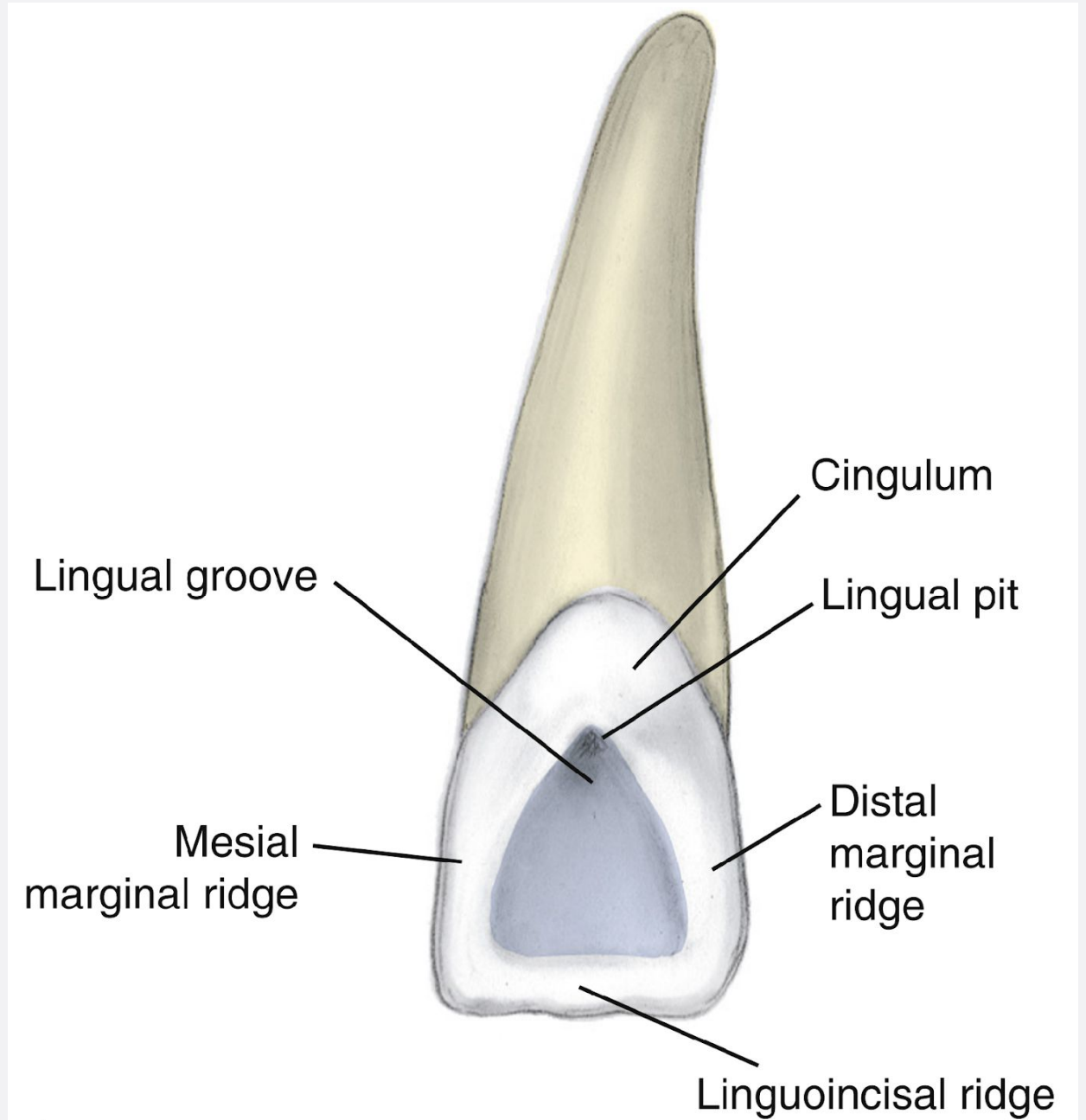
- Long, slender root

Root is about 1.5 times the length of crown.

Root apex pointed with distal taper at lower 1/3rd .



Lingual Surface



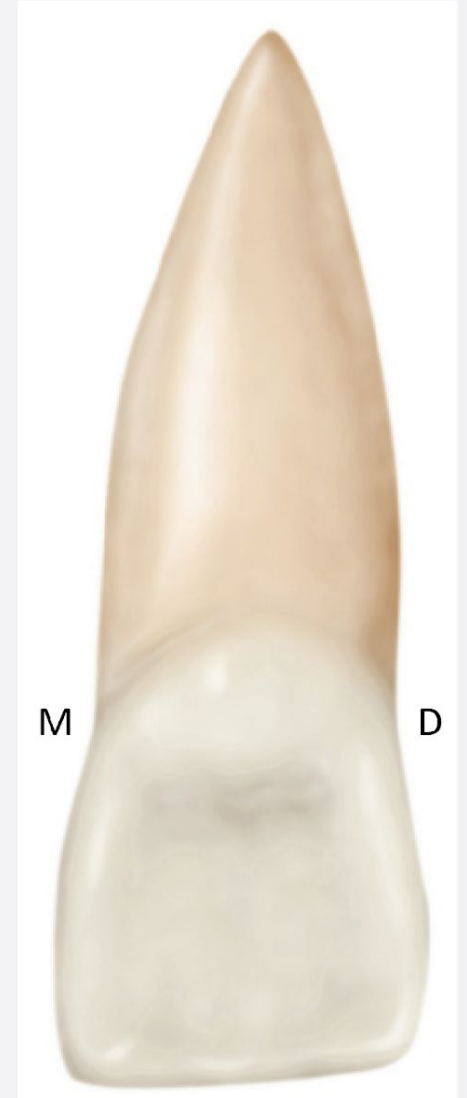
Lingual Aspect: Marginal Ridges

Corresponding/similar to
maxillary central incisor but
more pronounced

- **Mesial and distal marginal ridges are well marked / Prominent and well defined than maxillary central incisor**



Maxillary lateral incisor



Maxillary central incisor

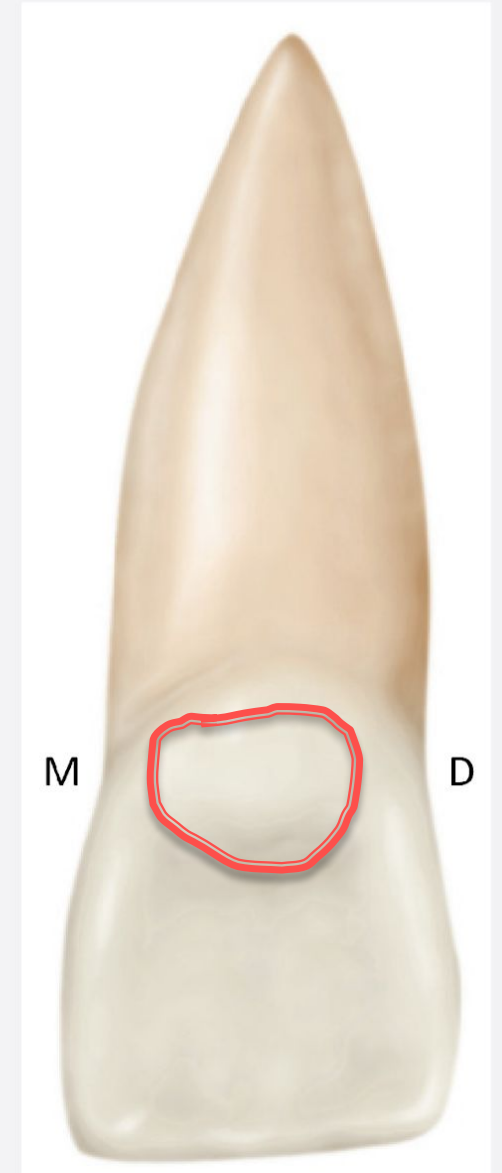
Lingual Aspect: Cingulum

Corresponding To Maxillary
Central Incisor

- Usually large and prominent.



Maxillary lateral incisor

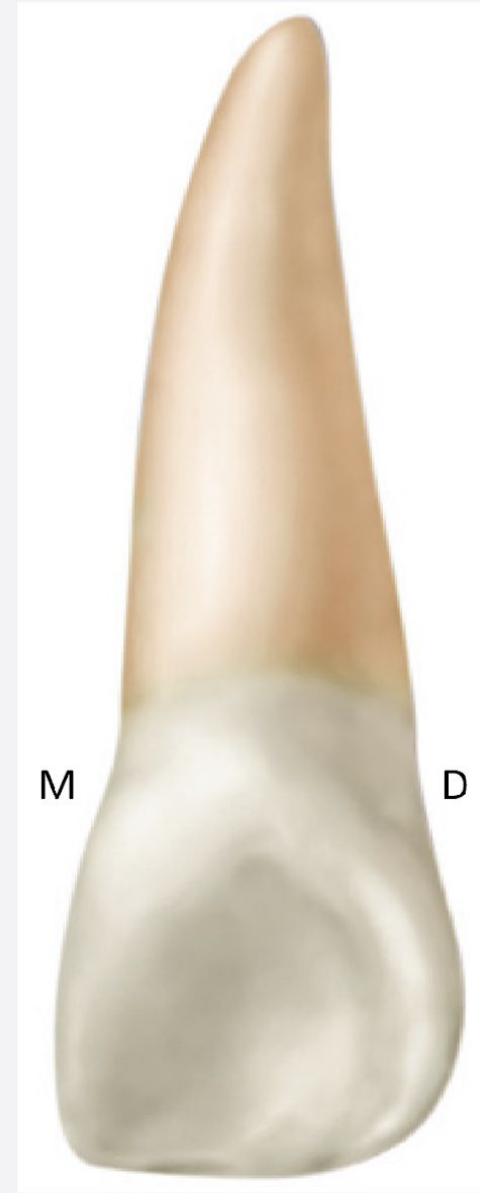


Maxillary central incisor

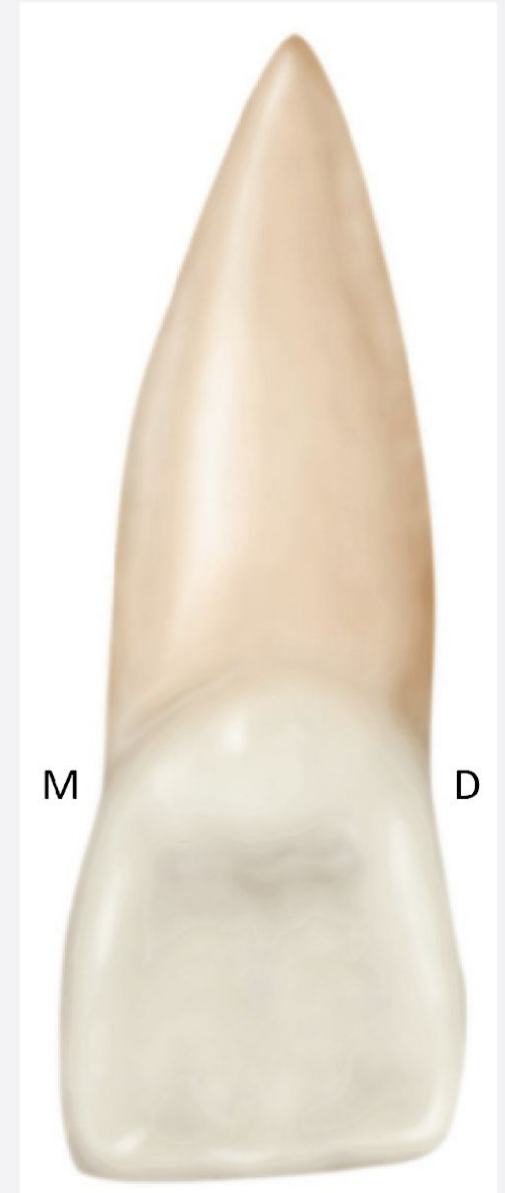
Lingual Aspect: Lingual Fossa

Corresponding To Maxillary Central Incisor

- **More concave, deep and circumscribed.**



Maxillary lateral incisor



Maxillary central incisor

Lingual Aspect: Linguo-Incisal Ridge

Corresponding To Maxillary Central Incisor

- **Well-developed.**



Maxillary lateral incisor



Maxillary central incisor

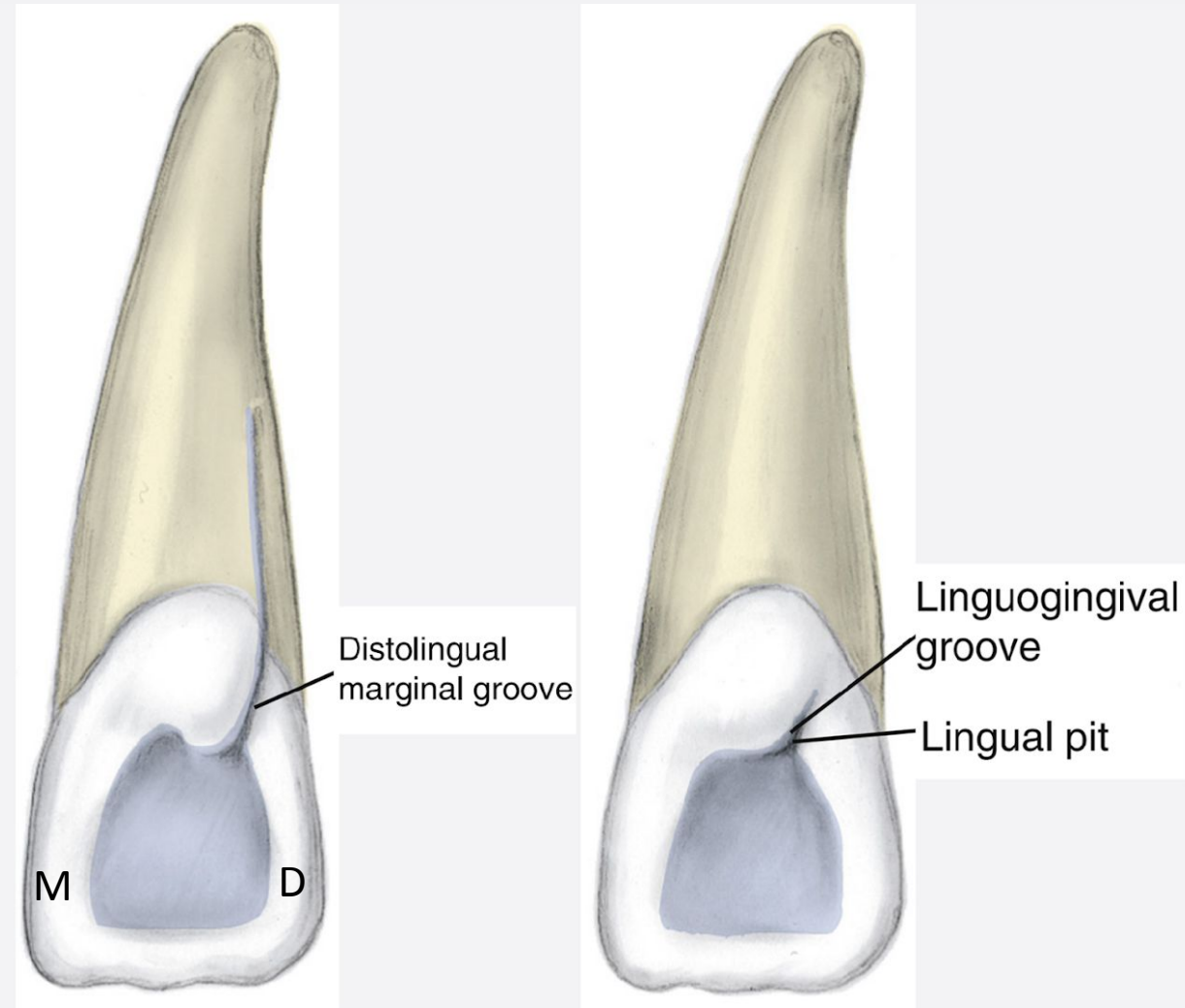
Lingual Aspect: Lingual Grooves & Variations

**A deep developmental groove:
Palato-radicular grooves**

Usually at the distal side of the
cingulum,

may extend up, on the root

for part or all its length.



Lingual Aspect: Lingual Grooves & Variations

Clinical significance:

Plaque retention → **deep periodontal pockets / localized periodontitis**

Often associated with **enamel pits**

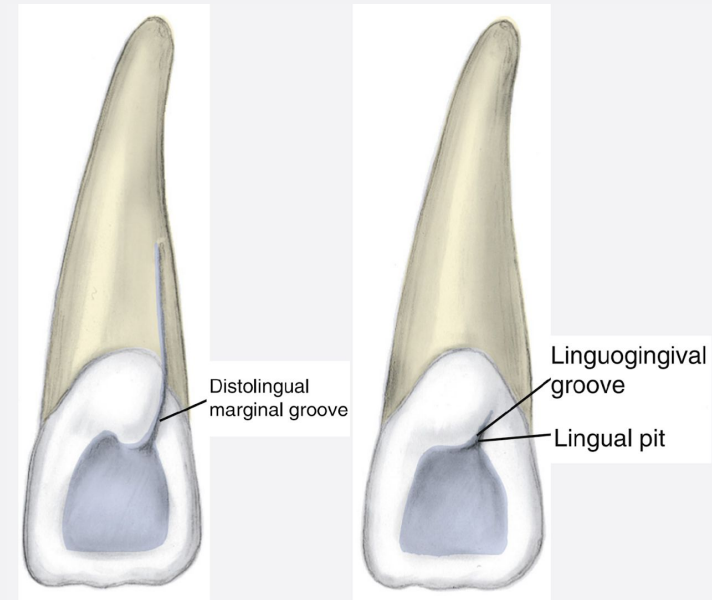


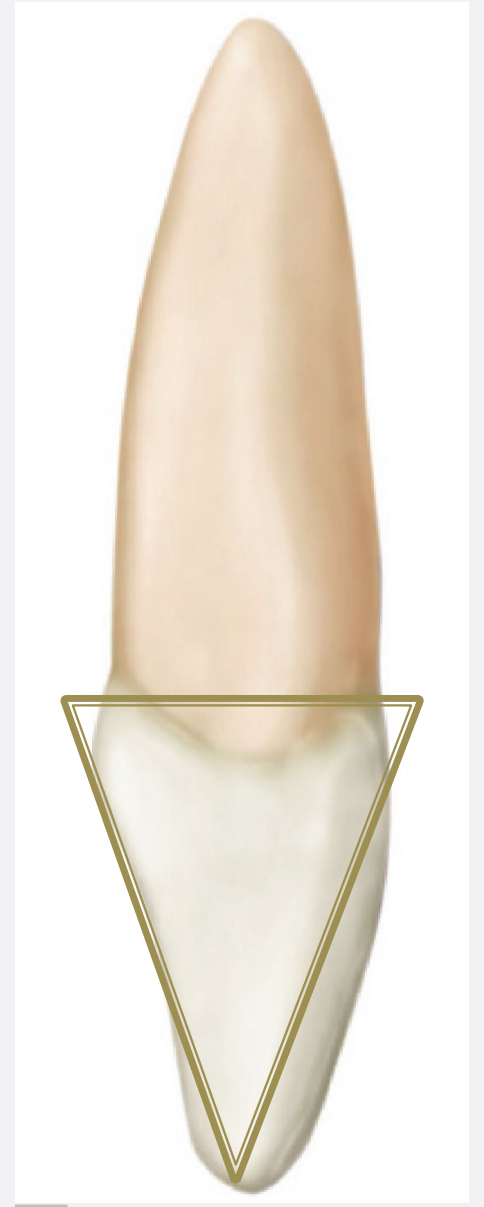
Figure 2. Probing of the pocket before surgery

Mesial And Distal Aspect Of Maxillary Lateral Incisor

**The crown outline is triangular on
a mesial and distal view**



Mesial Aspect



Distal Aspect

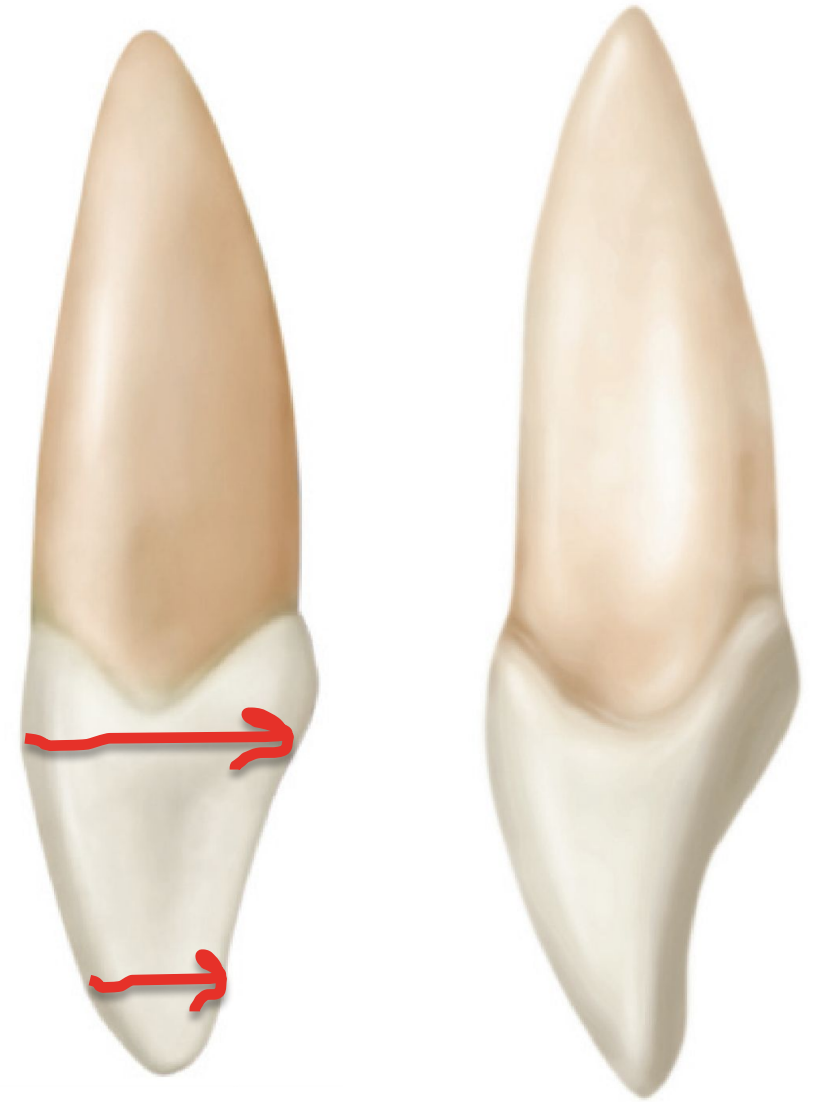
Mesial Aspect

Corresponding To Maxillary Central Incisor

The crown is shorter,

The labiolingual measurement of the crown and root is 1mm lesser.

The heavy development of the incisal ridge makes the incisal portion somewhat thicker.



Maxillary lateral incisor Maxillary central incisor

Mesial And Distal Aspect

Curvature Of The Cervical Line

Marked in the direction of the incisal ridge.

Actual extent of curvature is less than that of central incisor on mesial aspect.

The curvature of the cervical line is usually a less in depth on distal side than on the mesial side.



Mesial And Distal Aspect

The root appears as a tapered cone.

A bluntly rounded apical end.

A line drawn through the center of the root tends to bisect the incisal ridge of the crown.



Difference Between Mesial and Distal Aspect

Crown Thickness: Incisal third appears **thicker** (more labial surface visible)

Cervical Line: Curvature **~1 mm less** than mesial side

Developmental Groove: May occur on **distal crown extending onto root** (palatoradicular groove)

Incisal Aspect

Resembles a central incisor and canine except in size.

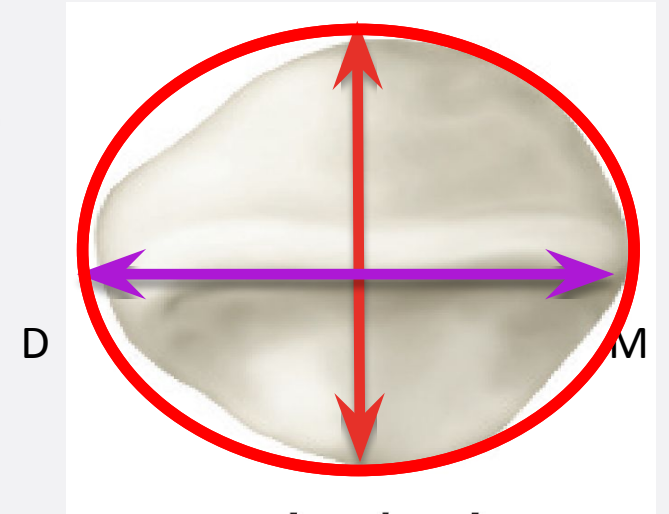
Corresponding to maxillary central incisor

The outline of the crown is round or oval, not triangular.

Exhibit more convexity labially and lingually.

The cingulum may be large, as is the incisal ridge.

The labiolingual dimension greater than mesiodistal dimension.



Maxillary lateral incisor



Maxillary central incisor

Variations & Anomalies

- **Peg Lateral:**
 - Small **conical / peg-shaped crown**
 - May have **malformed root**
- **Palatoradicular Groove:**
 - Deep groove from **lingual fossa extending onto root**
 - Can predispose to **periodontal problems**



Figure 2. Probing of the pocket before surgery

Variations & Anomalies

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 - Deep groove from **lingual fossa extending onto root**
 - Can predispose to **periodontal problems**
- **Tuberculate Cingulum:**
 - **Large pointed tubercle** on the cingulum
- **Root or Crown Malformation:**
 - **Twisted roots or distorted crown shape**
- **Agenesis (Missing Tooth):**
 - One of the **most commonly congenitally missing teeth**
 - Central incisor may **contact canine** if absent

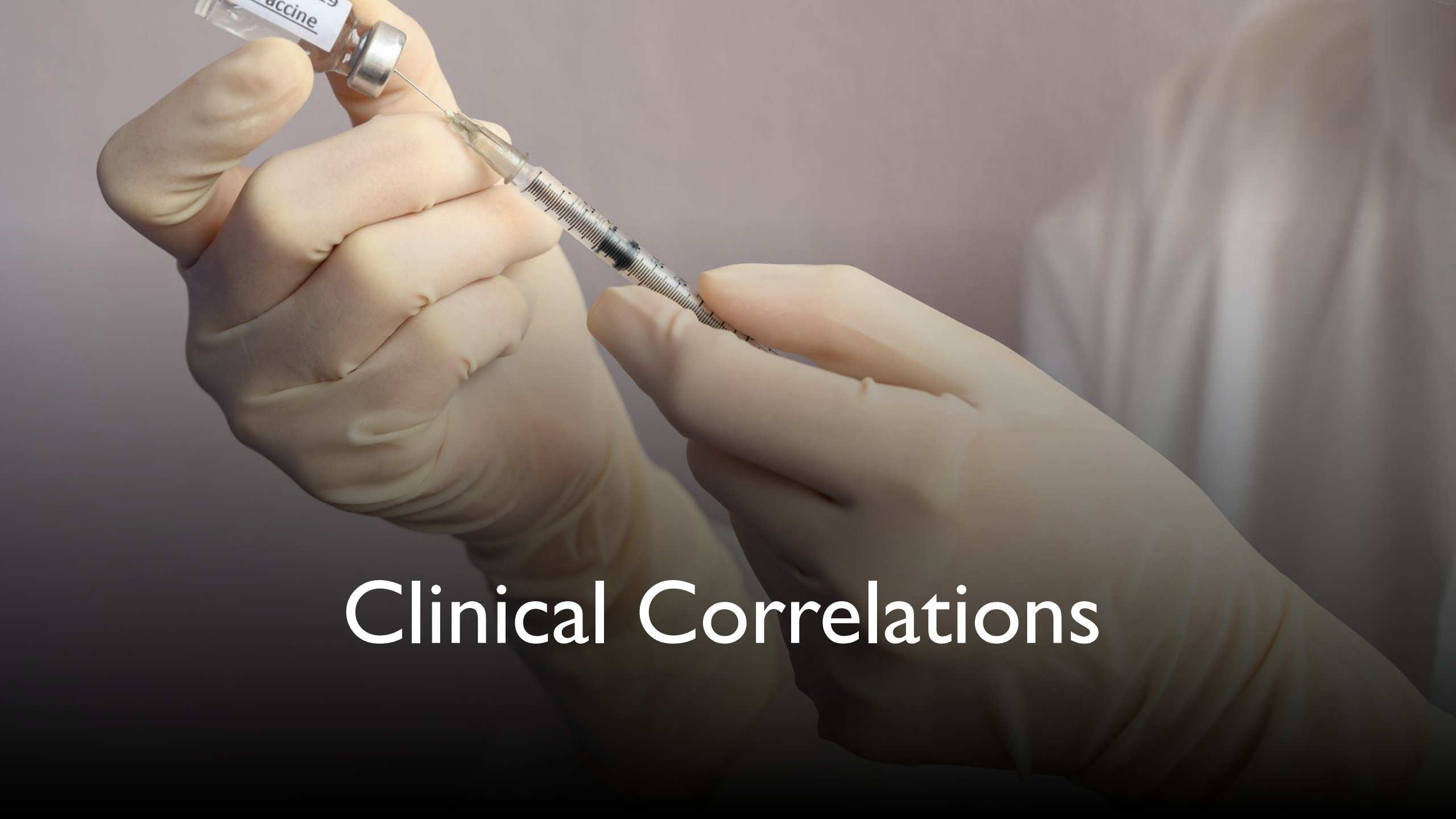
Variations & Anomalies

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	MAXILLARY CENTRAL INCISOR	MAXILLARY LATERAL INCISOR
General crown features	Incisal ridge, incisal angles, cingulum, marginal ridges, lingual fossa	
Specific crown features	Wider crown mesiodistally. Greater CEJ curve and height of contour. Pronounced distal offset cingulum Marginal ridges, with wide and deep lingual fossa	Greatest crown variation. More pronounced lingual surface, with centered cingulum and prominent marginal ridges
Height of contour	Cervical third	
Mesial contact	Incisal third	Junction of incisal and middle third
Distal contact	Junction of incisal and middle thirds	Middle third

	MAXILLARY CENTRAL INCISOR	MAXILLARY LATERAL INCISOR
Distinguish right from left	Sharper mesioincisal angle and rounder disto-incisal angle. More pronounced mesial CEJ curvature	
General root features	Single root	
Specific root features	Triangular in cross section. Overall conical shape. No proximal root concavities. Rounded apex	Oval in cross section. Same or longer than central but thinner. Overall conical shape. No proximal root concavities. Root curves distally, with sharp apex



Clinical Correlations

Maxillary lateral incisors vary in form more than any other tooth in the mouth except the third molar.

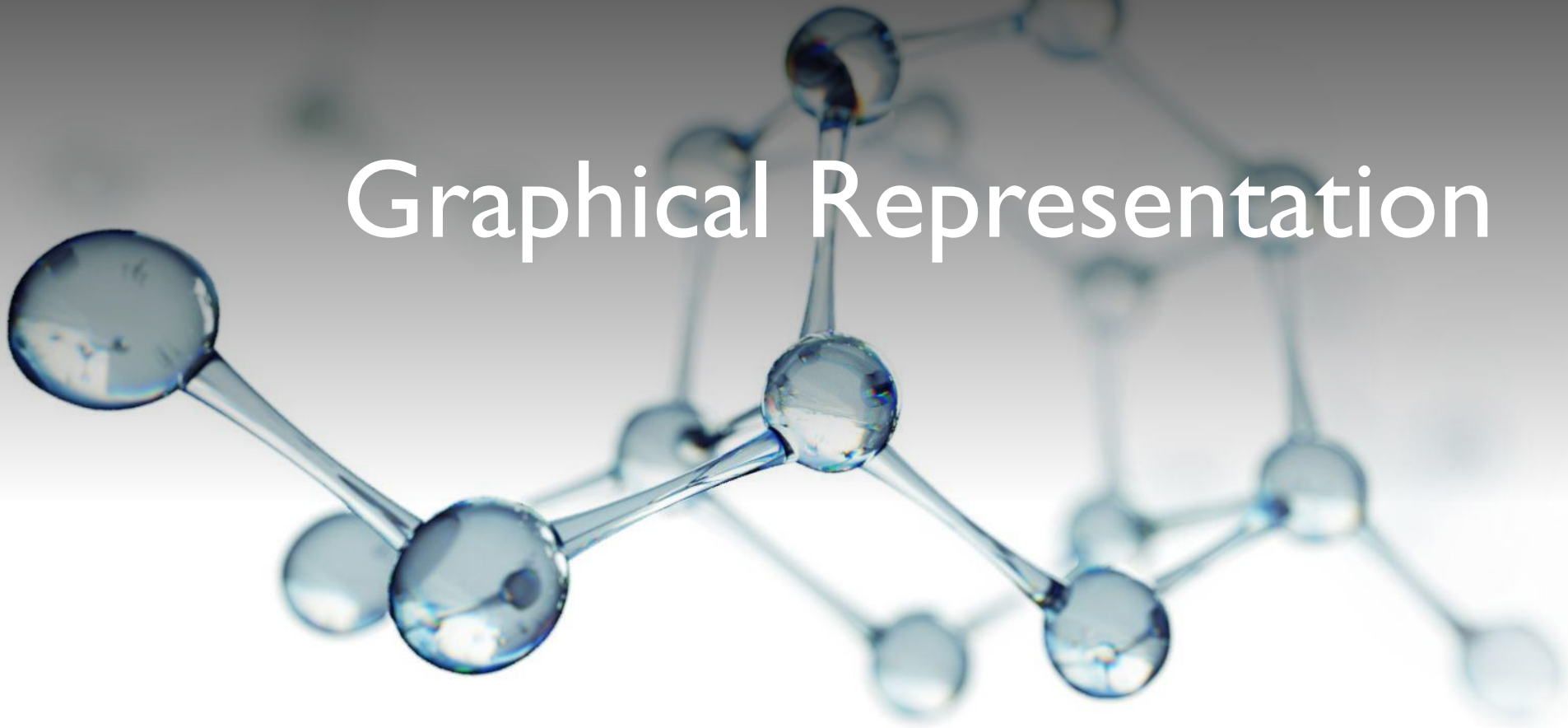
The presence of a palatogingival groove in maxillary incisors may be a predisposing factor in localized periodontal disease, also referred to as the palatoradicular groove.



**Maxillary lateral incisors
with a nondescript,
pointed form; such teeth
are called **peg-shaped
laterals**.**

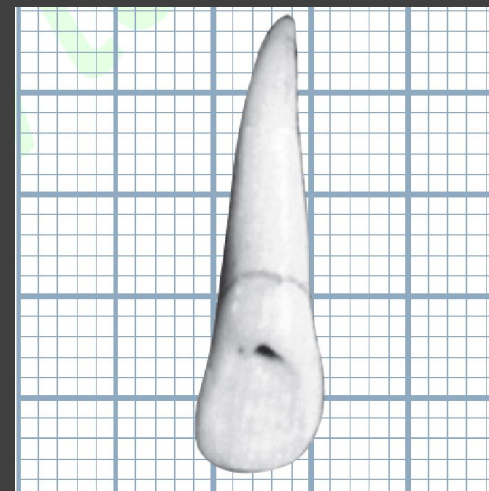


Graphical Representation

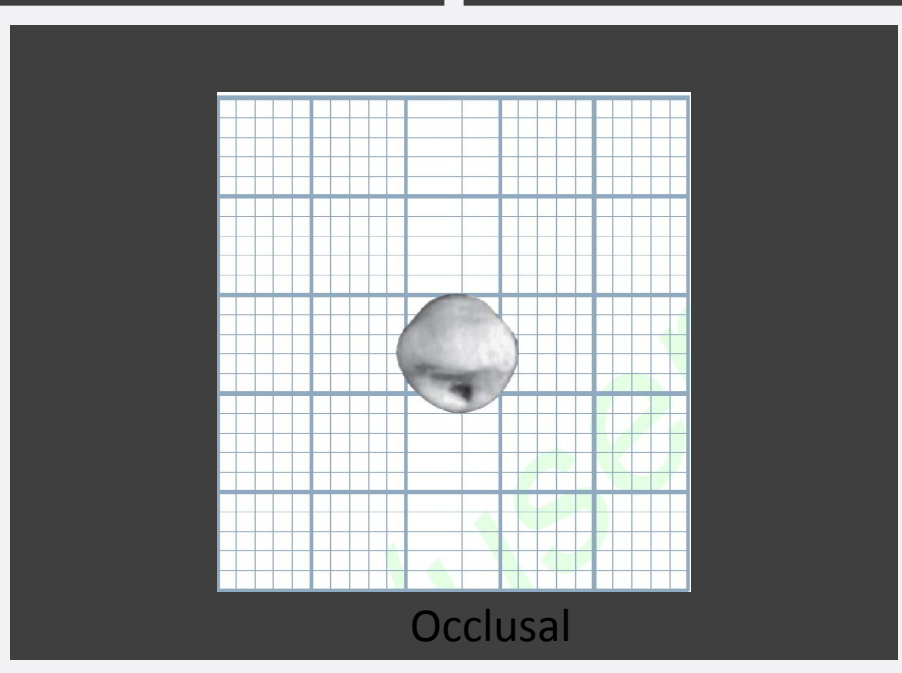




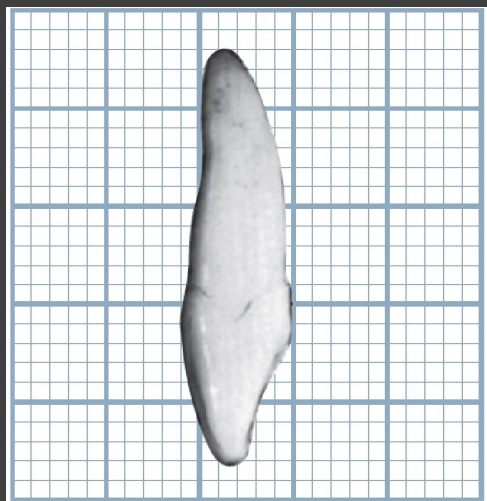
labial



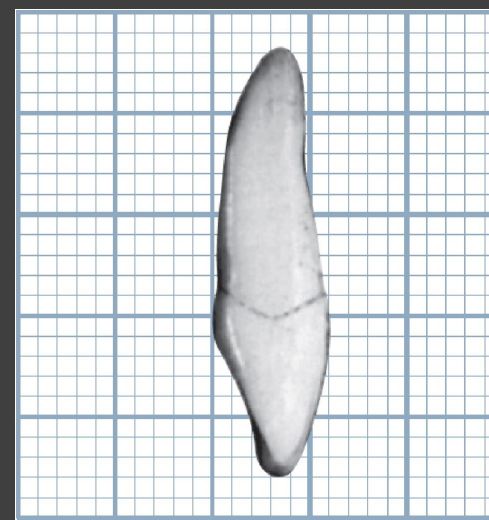
Lingual



Occlusal



Mesial



Distal

Thank you!

